



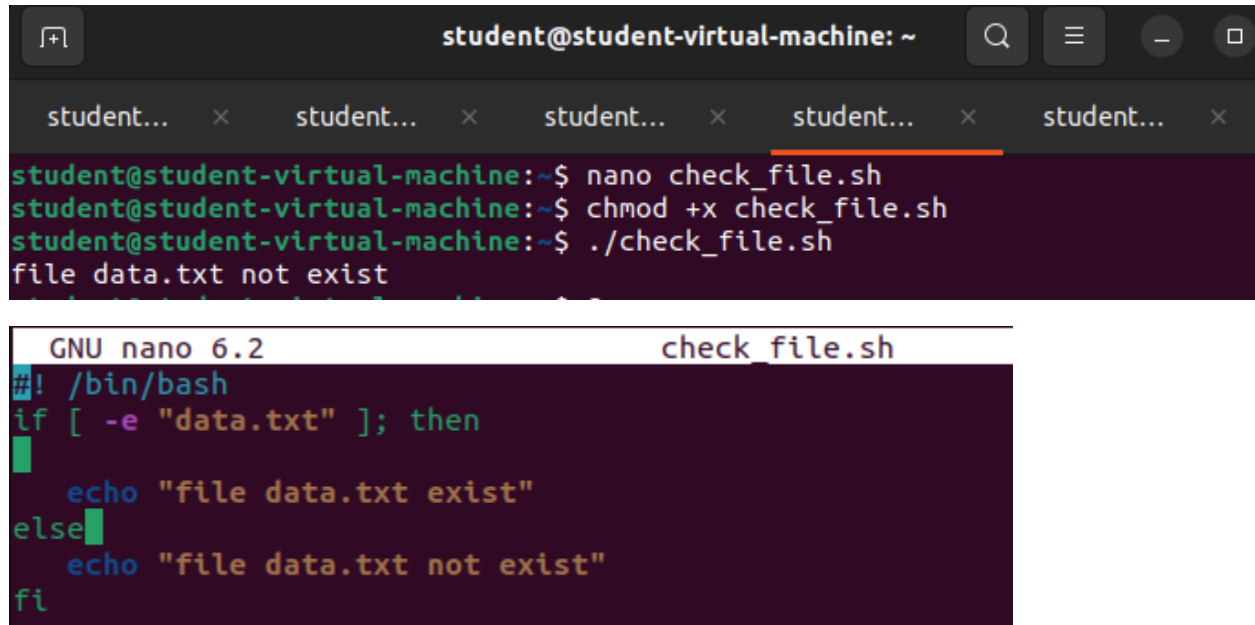
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Lab 10 :Operating system

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Task 1:



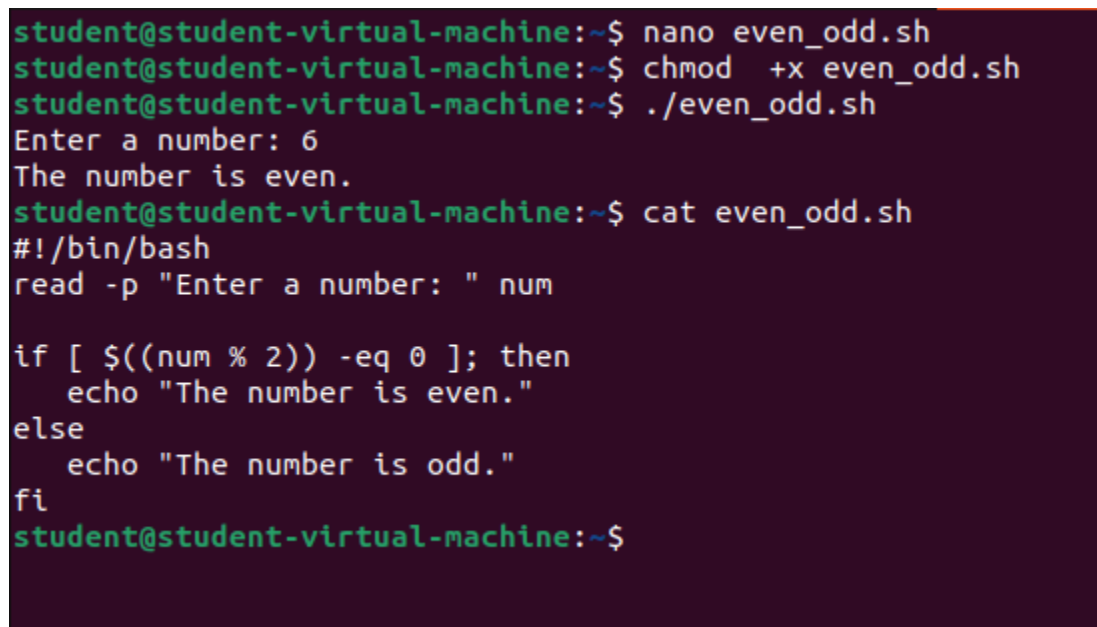
The image shows a terminal window titled 'student@student-virtual-machine: ~'. It contains a sequence of commands and their output:

```
student@student-virtual-machine:~$ nano check_file.sh
student@student-virtual-machine:~$ chmod +x check_file.sh
student@student-virtual-machine:~$ ./check_file.sh
file data.txt not exist
```

Below the terminal window, the content of the file 'check_file.sh' is displayed in a code editor (GNU nano 6.2):

```
#!/bin/bash
if [ -e "data.txt" ]; then
    echo "file data.txt exist"
else
    echo "file data.txt not exist"
fi
```

Task 2:



The image shows a terminal window titled 'student@student-virtual-machine: ~'. It contains a sequence of commands and their output:

```
student@student-virtual-machine:~$ nano even_odd.sh
student@student-virtual-machine:~$ chmod +x even_odd.sh
student@student-virtual-machine:~$ ./even_odd.sh
Enter a number: 6
The number is even.
student@student-virtual-machine:~$ cat even_odd.sh
#!/bin/bash
read -p "Enter a number: " num

if [ $((num % 2)) -eq 0 ]; then
    echo "The number is even."
else
    echo "The number is odd."
fi
student@student-virtual-machine:~$
```

Task 3:

```
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student@student-virtual-machine:~$ nano grade.sh  
student@student-virtual-machine:~$ chmod +x grade.sh  
student@student-virtual-machine:~$ ./grade.sh  
Enter your score (0-100): 80  
Grade: B  
student@student-virtual-machine:~$ 90  
90: command not found  
student@student-virtual-machine:~$ cat grade.sh  
#!/bin/bash  
read -p "Enter your score (0-100): " score  
  
if [ $score -ge 90 ]; then  
    echo "Grade: A"  
elif [ $score -ge 80 ]; then  
    echo "Grade: B"  
elif [ $score -ge 70 ]; then  
    echo "Grade: C"  
elif [ $score -ge 60 ]; then  
    echo "Grade: D"  
else  
    echo "Grade: F"  
fi
```

TASK4:

```
student@student-virtual-machine:~$ nano calculator.sh
student@student-virtual-machine:~$ chmod +x calculator.sh
student@student-virtual-machine:~$ ./calculator.sh
Enter first number: 6
Enter second number: 2
Enter operator (+, -, *, /): *
Result: 12
student@student-virtual-machine:~$ ./calculator.sh
Enter first number: 5
Enter second number: 2
Enter operator (+, -, *, /): -
Result: 3
student@student-virtual-machine:~$ cat calculator.sh
#!/bin/bash
read -p "Enter first number: " num1
read -p "Enter second number: " num2
read -p "Enter operator (+, -, *, /): " op

if [ "$op" = "+" ]; then
    result=$((num1 + num2))
    echo "Result: $result"
elif [ "$op" = "-" ]; then
    result=$((num1 - num2))
    echo "Result: $result"
```

```
    echo "Result: $result"
elif [ "$op" = "*" ]; then
    result=$((num1 * num2))
    echo "Result: $result"
elif [ "$op" = "/" ]; then
    result=$((num1 / num2))
    echo "Result: $result"
else
    echo "Invalid operator!"
fi
student@student-virtual-machine:~$
```